
★ SPECIAL SECTION • CONCLUSION

Data modeling stopped being intimidating somewhere in this Refcard. Chapter 1 gave you the vocabulary - OLTP versus OLAP, schema-on-write versus schema-on-read, why every dimension needs a surrogate key.

Chapter 2 introduced Kimball's four-step design, and grain became the single most consequential decision you now know how to make. Chapter 3 solved the biggest operational problem in the field - how to preserve history when the world changes - by making effective_**from, effective_to**, and iscurrent second nature.

Chapter 4 zoomed out to schema patterns and pitfalls. Chapter 5 placed your models in their modern homes (warehouse, lake, lakehouse) and taught you the vocabulary of Delta, Iceberg, and the medallion architecture.

Chapter 6 crossed into the ML world with feature stores, point-in-time correctness, and train-serve skew. Then Lab 1 delivered a complete runnable retail warehouse for Mango Mart, and Lab 2 answered ten real business questions against it with production-shaped SQL.\n\nYou now hold the same mental model that senior data engineers apply on their first week at a new company.

Given a domain you've never seen - a hospital, a logistics fleet, a streaming service, a food-delivery app - you can conduct the design conversation, declare grain in a single sentence, choose the right SCD type per attribute, sketch the star schema on a whiteboard, and translate it into runnable DDL by the end of the day. You know when a warehouse is enough and when to reach for a lakehouse.

You know when a feature store earns its complexity and when it doesn't. Every KPI query in Lab 2 was a variation on one shape: SELECT dim-attributes and aggregated-measures FROM fact JOIN dims WHERE filters GROUP BY dim-attributes - a shape that answers virtually any business question with minor recombination.

Where to go from here: the fastest returns come from building a real project. Pick a dataset that interests you - public transit records, your own personal budget, a game's leaderboard, an e-commerce sample on Kaggle - and design its star schema end-to-end following Lab 1's six steps.

Deploy it to a free-tier Snowflake or BigQuery account. Load the data with dbt, the tool most professional teams use to manage the SCD Type 2 and materialization patterns you just learned.

Write ten KPI queries against your model, following Lab 2's shape. Publish the repo - it's the strongest possible portfolio artifact for a data-engineering role.

Then keep going: read Kimball's original *The Data Warehouse Toolkit*, contribute to an open-source data project, and follow the practitioners publishing about warehouse modernization on modern data stacks. You've built the foundation.

Everything from here is repetition, refinement, and shipping. Go build something.